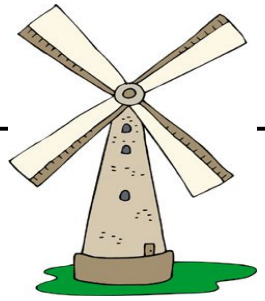
Name: _____

Deep Thoughts on Matter



1. Since you can pour sand into a cup, why isn't it a liquid?

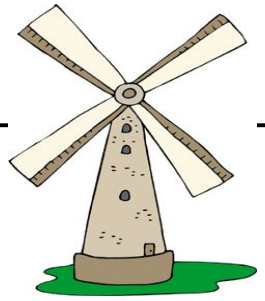
2. You can't usually see gases in the air. How can you observe gases without seeing them?

3. Can you feel water in the air? Explain.

4. Does air have mass? Explain.

ANSWER KEY

Deep Thoughts on Matter



1. Since you can pour sand into a cup, why isn't it a liquid?

Each particle of sand has a definite shape and volume. When billions of grains of sand are put together, they can be poured because they are so small.

2. You can't usually see gases in the air. How can you observe gases without seeing them?

You can feel the wind blow. You can observe air movement, such as a waving flag or a turning windmill. You can also observe temperature and humidity changes in the air. You can also smell many types of gases.

3. Can you feel water in the air? Explain.

Yes, you can feel humidity in the air. You would notice that the air is different on a damp or muggy day then on a dry day. Some people can even feel it in their joints. Some people might think things feel more "sticky" in a very humid environment.

4. Does air have mass? Explain.

Yes, air has mass. Since it is lighter than liquids and solids, it tends to rise above them. One easy experiment to test this would be to find the mass of an empty balloon and an inflated balloon. The inflated balloon should be heavier. Also, you might notice that an empty air tank (propane, for example) weighs less than a full air tank.